How Well Can an Ensemble of Large Language Models Replicate a Human Review of Full-Length Transplant Related Manuscripts Using the USRDS?

Anagha Sreenivasan[1] , Ladan Golestaneh[2] , Vagish Hemmige[3]
1 University of Texas at Austin, Austin, TX,
2 Yale University School of Medicine, New Haven, CT
3 Montefiore Medical Center, Bronx, NY

BACKGROUND

- The United States Renal Data System (USRDS) merges individual claims data from Medicare and transplant data from UNOS.
- Accordingly, the USRDS is an important resource for studying kidney disease, transplantation, and related outcomes, but usage is complicated by complex structure (400+ files) and size (1TB +).
- We previously performed a manual review of manuscripts obtained by a PubMed search to examine how the various components of the USDRS were used by researchers.
- Large Language Models (LLMS) may automate this type of review, but the accuracy of such a process is unknown.

OBJECTIVE

Evaluate the efficiency of LLMs in reviewing USRDS transplant manuscripts.

METHODS

- We manually reviewed a sample of USRDS papers and recorded files types, programming languages, and elements used.
- We built an R-based pipeline from raw PDF to three LLMs for data abstraction.
- We compared each LLM's answers to consensus human answer.
- Cohen's K was used to quantify chance-corrected agreement between each LLM and human reviewers for every extracted variable.

RESULTS

Figure 1. Comparing Human Review and LLM Extraction

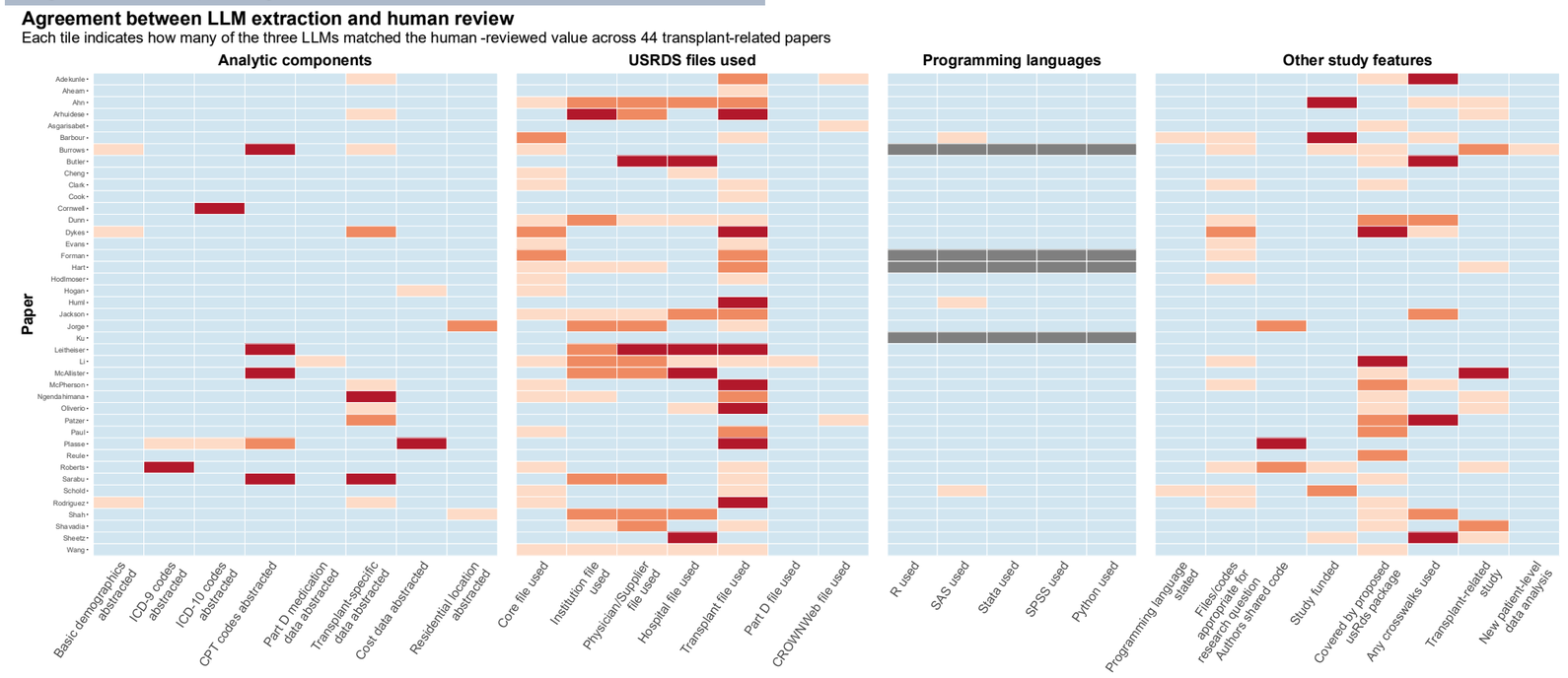


Table 1. Runtime of LLMs to Process the Full Dataset

Runtime of LLMs Used in the Analysis

Wall-clock time in minutes to process the full dataset

Provider (Company)	Model	Time (minutes)
ChatGPT (OpenAI)	Gpt 5.1	165
Claude (Anthropic)	Claude Sonnet 4.5	62
Gemini (Google)	Gemini 2.5 Flash	672

Table 2. Cohen's K Analysis

Cohen's κ: LLM vs Human Across All Variables

Dataset of 44 transplant-related papers

Variable	ChatGPT	Claude	Gemini
Components of USRDS used			
Basic demographics abstracted	0.000	-[1]	-[1]
Cost data abstracted	0.482	0.656	0.656
CPT codes abstracted	0.699	0.638	0.638
ICD-10 codes abstracted	0.920	0.920	0.848
ICD-9 codes abstracted	0.955	0.955	0.909
Part D medication data abstracted	0.788	1.000	1.000
Residential location abstracted	0.950	0.952	0.904
Transplant-specific data abstracted	0.450	0.135	−0.038
Files used			
Core file used	0.000	0.000	0.000
CROWNWeb file used	1.000	1.000	0.692
Hospital file used	0.294	0.542	0.267
Institution file used	0.364	0.636	0.773
Part D file used	0.788	1.000	1.000
Physician/Supplier file used	0.343	0.629	0.770
Transplant file used	0.065	0.078	0.276
Programming languages			
Python used	1.000	1.000	1.000
R used	1.000	1.000	1.000
SAS used	0.894	0.946	1.000
SPSS used	1.000	1.000	1.000
Stata used	1.000	1.000	1.000
USRDS files explicitly stated	-[1]	-[1]	-[1]
Authors shared code	−0.031	0.656	−0.031
Programming language stated	0.776	1.000	1.000
Files/codes appropriate for research question	0.000	0.000	-[1]
Transplant-related study	0.000	0.000	0.000
New patient-level data analysis	0.000	-[1]	-[1]
Study funded	0.841	0.791	0.784
Any crosswalks used	0.391	0.310	0.438
Covered by proposed usRds package	0.000	0.000	0.000

[1] κ not estimable, usually due to insufficient variation or no usable paired observations.

Cohen's κ legend

Agreement	Range
Negative	< 0
Exact zero	0.000
Slight	0.001–0.20
Fair	0.21–0.40
Moderate	0.41–0.60
Substantial	0.61–0.80
Almost perfect	0.81–1.00
Not estimable	-

CONCLUSION/NEXT STEPS

- The USRDS remains underutilized as a resource to improve post-transplant outcomes for patients.
- Additionally, large language models have the potential to be a valuable resource for analyzing transplant papers.
- Future work should explore the improvement of LLM prompts and the use of more advanced models in reviewing literature.